Business Fluctuations and Firm-Level Responses to Economic Crises

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GitHub Repository Link: https://github.com/RavinduWeer/Business-Fluctuations-and-Firm-Level-Responses-to-Economic-Crises.git

1. Cross-Country Business Cycle Analysis

1.1 Introduction

This analysis examines the business cycle fluctuations of the United States, France and Australia from 2005 to 2024. Using quarterly data obtained from FRED, we analyze key macroeconomic variables: Consumer Price Index (CPI), 10-Year Government Bond Yields (LongTermBond), Unemployment and Gross Domestic Product (GDP).

All data was transformed to a quarterly frequency and detrended using a Hodrick-Prescott (HP) filter (λ=1600) to isolate the cyclical components. This report interprets the findings on volatility, correlation with output and overall cyclical dynamics based on the graphical output generated.

1.2 Volatility Analysis

The volatility of an economy which is measured by the standard deviation of its detrended variables, provides insight into the magnitude of its cyclical fluctuations. The analysis presented in Figure 01 reveals significant differences between the three countries.

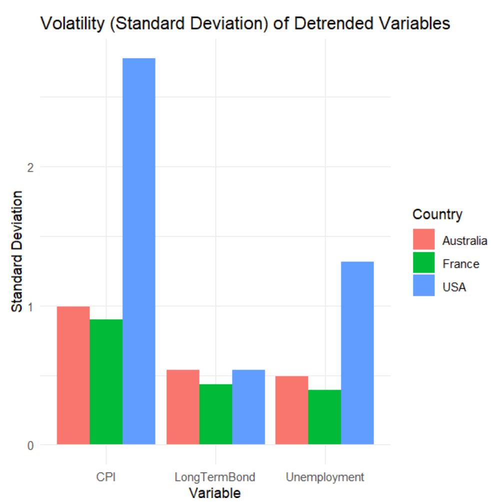

Figure 01: Volatility (Standard Deviation) of Detrended variables

Prices (CPI): The most striking difference is in price volatility. The USA exhibits exceptionally high CPI volatility (Std. Dev. approximately 2.8), nearly three times that of Australia (approximately 1.0) and

France (approximately 0.9). This suggests that price-level shocks and subsequent fluctuations are a much more prominent feature of the US business cycle.

Labor Market (Unemployment): A similar, though less extreme, pattern appears in the labor market. The **USA** again shows the highest volatility in its unemployment cycle (Std. Dev. approximately 1.3). This is more than triple the volatility seen in France (approximately 0.3) and Australia (approximately 0.4). This points to a US labor market with much larger cyclical swings in joblessness compared to the more stable French and Australian labor markets.

Financial Markets (LongTermBond): In contrast, long-term bond yields show relatively uniform and low volatility across all three nations, with France (Std. Dev. approximately 0.4) being slightly more stable than the USA (approximately 0.5) and Australia (approximately 0.5). This suggests that long-term financial market cycles are more globally integrated and less subject to the specific domestic instabilities seen in US prices and labor.

The United States economy, within this period, is characterized by significantly more volatile price and labor market cycles. France and Australia appear to have more muted and stable domestic business cycles.

1.3 Correlation with Output (GDP)

This section analyzes the cyclical relationships of variables with GDP (output), as detailed in Figure 02. This determines if variables are procyclical (move with GDP), countercyclical (move against GDP), or acyclical (no clear relationship).

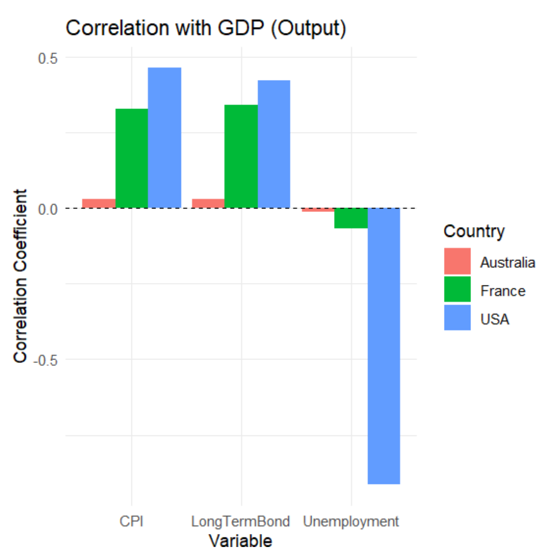

Figure 02: Correlation with GDP (Output)

- **United States (Textbook Cycle):** The US follows a "textbook" business cycle pattern.

CPI and LongTermBond are both clearly procyclical (Correlation approximately +0.48 and +0.45, respectively). When the economy (GDP) expands, inflation and long-term interest rates tend to rise.

Unemployment is strongly countercyclical (Correlation approximately -0.8), indicating that as GDP falls, unemployment robustly rises and vice-versa.

- **France (Muted Cycle):** France shows a similar, but more muted, pattern to the US.

CPI and LongTermBond are also procyclical (Correlation approximately +0.35 and +0.38, respectively).

However, Unemployment is only very weakly countercyclical (Correlation approximately -0.1). This suggests a much weaker relationship between output and the labor market compared to the US.

- **Australia (Acyclical Anomaly):** Australia presents a stark contrast to the other two nations.

CPI, LongTermBond and Unemployment are all acyclical, with correlation coefficients very close to zero. This is a significant finding, suggesting that Australia's output fluctuations (perhaps driven by external factors like commodity prices) are largely disconnected from its domestic inflation, long-term interest rates and labor market cycles.

1.3 Analysis of Cyclical Dynamics

Figure 03 provides a direct visual comparison of the cycles over time, confirming the findings from the first two charts.

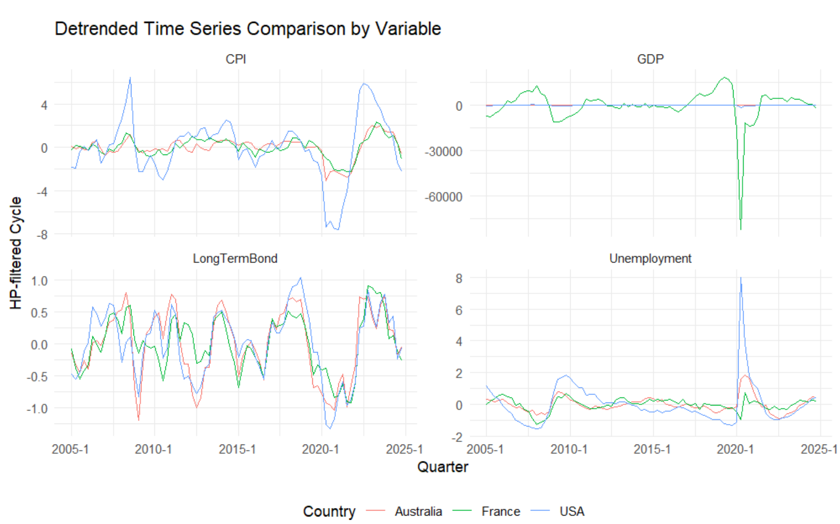

Figure 03: Detrended Time Series Comparison by Variable

The "spikiness" of the blue line (USA) in the CPI and Unemployment panels visually confirms the high volatility identified in Figure 1. The French (green) and Australian (red) lines are comparatively smooth. The time series plots show how the countries respond to major global events.

Global Financial Crisis (c. 2008-200G): The GFC is visible as a significant event in all three economies. The GDP panel shows a clear dip for all three countries. The Unemployment Panel shows a corresponding large spike for the USA, but a much smaller reaction in France and Australia, reinforcing their lower labor market volatility.

COVID-1G Pandemic (c. 2020): The 2020 shock shows highly divergent impacts. France's GDP (green line) experienced a catastrophic and immediate drop, far exceeding that of the US or Australia. Simultaneously, US Unemployment (blue line) saw an unprecedented spike, an order of magnitude larger than any other event in the series. US CPI also shows a major deflationary shock followed by a massive inflationary spike post-2020, which is absent in the other countries.

1.4 Conclusion

The business cycles of the United States, France and Australia, while subject to the same global shocks, are structurally different.

The United States experiences a high-volatility cycle, particularly in prices and unemployment, that follows a classic procyclical pattern for inflation/interest rates and a strongly countercyclical pattern for unemployment.
France has a much more stable (low-volatility) economy. It shares the procyclical nature of the US for inflation and bonds, but its labor market is less responsive to output changes.

Australia stands apart. It exhibits low volatility like France, but its domestic inflation, financial markets and labor markets appear almost entirely acyclical, suggesting its output cycle is driven by different factors (e.g., global commodity demand) than the other two advanced economies.

2 Event Study of the GFC and COVID-1G Recessions in the USA

2.1 Introduction

This report analyzes and contrasts two distinct economic crises in the United States: the Global Financial Crisis (GFC) (2007-2009) and the COVID-19 Recession (2020). The objective is to track macroeconomic dynamics, identify the shocks and propagation mechanisms and compare the severity and recovery of each event. The analysis uses quarterly US data from 2000-2023 for Real GDP, the Unemployment Rate, the SCP 500 Index and the Consumer Price Index (CPI).

2.2 Macroeconomic Dynamics: Before, During and After

An initial look at the macroeconomic variables in their levels with recessions shaded provides a clear overview of the long-term trends and the immediate impact of each crisis.

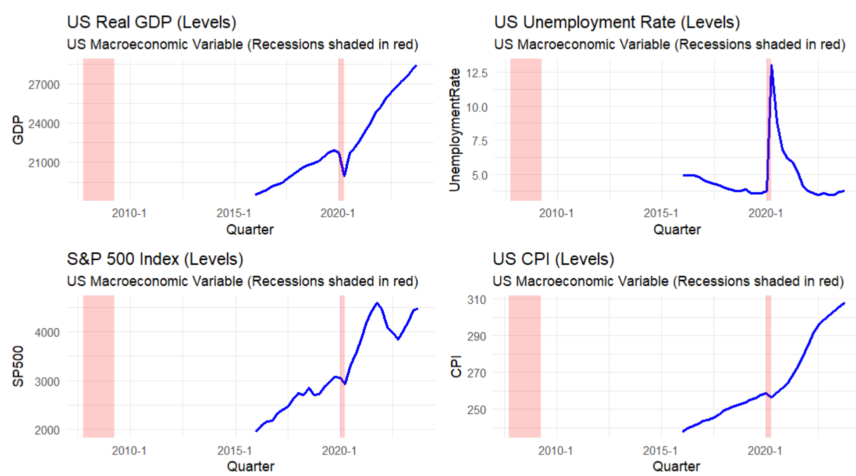

Figure 04: Variation of Real GDP (Levels), Unemployment Rate (Levels), SCP 500 Index (Levels) and CPI (Levels) of USA

Real GDP: Both recessions are visible as sharp, temporary deviations from a clear upward trend in Real GDP. The GFC dip (c. 2008-2009) appears as a prolonged "U-shape," where the economy contracted and took several quarters to regain its prior peak. In stark contrast, the COVID-19 dip (c. 2020) is a sudden, deep "V-shape," with a historic drop followed by a rapid recovery.

Unemployment Rate: The GFC is characterized by a "slow-burn" increase in unemployment, which starts during the recession and *continues to rise* even after the recession officially ended, peaking around 2010. The COVID-19 event shows a single, unprecedented spike in the unemployment rate, which peaks and begins to fall almost immediately.

SsP 500 Index: The GFC saw a deep, protracted bear market, with the SCP 500 index falling consistently throughout 2008 and not recovering its peak for several years. The COVID-19 crash was an extremely sharp, but brief, market panic, followed by a remarkably fast and strong bull market recovery.

CPI: The price level shows a disinflationary (or mildly deflationary) dip during the GFC, corresponding to the collapse in aggregate demand. The COVID-19 event shows an initial dip followed by a dramatic and sustained inflationary surge, which began in 2021.
When viewing these variables as growth rates or quarterly changes (Figure 05: Growth Rates and Changes in Key Macro Variables), the difference in *shock* becomes even clearer.

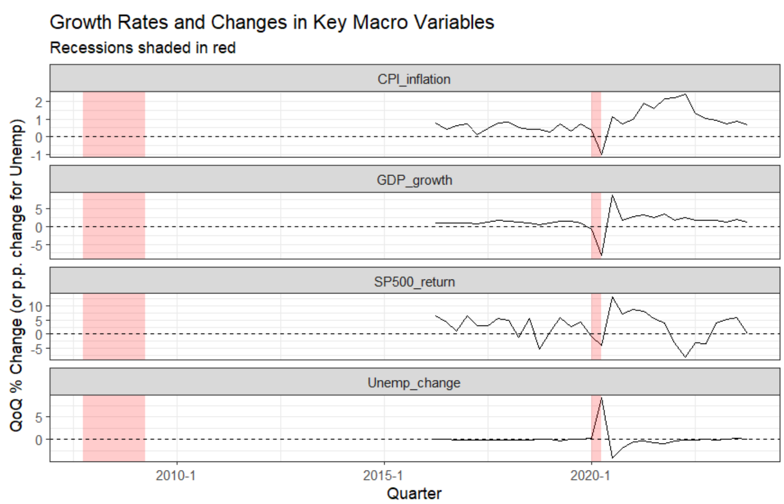

Figure 05: Growth Rates and Changes in Key Macro Variables

The COVID-19 recession was an event of unparalleled magnitude in a single quarter. GDP_growth plummeted to nearly -9% in one quarter, far exceeding the GFC's worst quarter of approximately -2.5%. Similarly, the Unemployment change (quarterly change in unemployment) saw a single-quarter spike of over 5 percentage points, an event with no precedent in the GFC, which was characterized by *persistent* but *smaller* quarterly increases in unemployment.

2.3 Comparative Analysis: Severity, Duration and Recovery

The analysis of the GFC as a protracted "U-shape" and COVID-19 as a "V-shape" is confirmed by Figure 06. These plots normalize each variable to 100 at the start of the recession (T=0) to compare their paths.

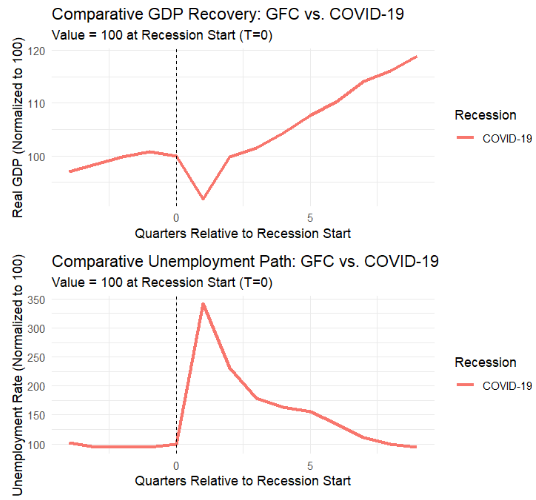

Figure 06: Comparative GDP Recovery and Comparative Unemployment Path

GDP Path (COVID-1G): The visible line for the COVID-19 recession shows Real GDP falling to approximately 92% of its pre-recession level in just one quarter. However, the recovery was equally swift. By T=2 (two quarters after the start), it had recovered most of the loss and by T=4, it was already 8% above its pre-recession level. This confirms the "V-shape" recovery in output.

Unemployment Path (COVID-1G): The unemployment path is even more dramatic. The unemployment rate exploded to nearly 350% of its starting value (a 3.5x increase) in one quarter. The subsequent recovery, while rapid, was slower than the GDP recovery. It took approximately 8 quarters (T=8) for the unemployment rate to return to its pre-recession level.

GFC (Impulse s Propagation): A financial shock (impulse) that began in the housing and banking sectors. It propagated slowly through the economy via credit constraints, deleveraging and a collapse in confidence, leading to a protracted recession (a "U-shape") and a "jobless recovery" where unemployment remained high for years.

COVID-1G (Impulse s Propagation): An external, non-economic shock (impulse) (a public health crisis). This shock acted as both a massive, immediate supply shock (lockdowns) and demand shock (fear, social distancing). It propagated instantly across all sectors. The recovery was fueled by unprecedented fiscal and monetary policy, leading to a sharp "V-shape" rebound.
The Figure: GDP Growth vs Change in Unemployment Rate plot, which visualizes Okun's Law, highlights this difference. The GFC quarters (not explicitly colored but part of the "Non-Recession" pink dots) fall along the standard negative trend. The COVID-19 quarters (blue dots) are extreme outliers, demonstrating a simultaneous, off-the-charts collapse in GDP and spike in

unemployment that exists far outside the normal business cycle dynamics.

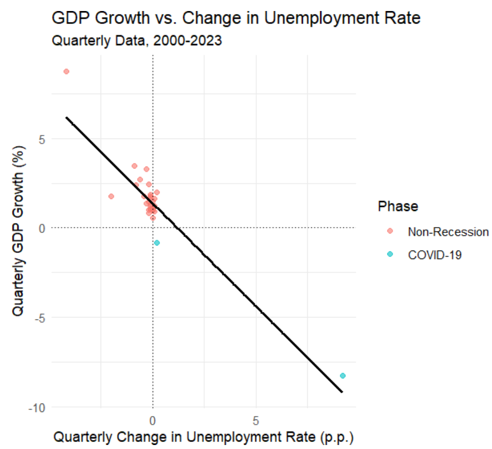

Figure 7: GDP Growth vs Change in Unemployment Rate

2.4 Shock Propagation and Amplification Mechanisms

To understand how these shocks spread, we analyze the underlying relationships between the variables. The Figure: Correlation Heatmap reveals the typical relationships in the US economy.

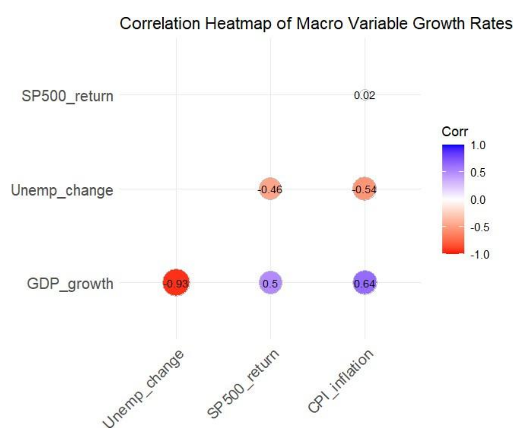

Figure 8: Correlation Heatmap of Macro Variable Growth Rates

GDP_growth vs. Unemp_change (-0.63): A near-perfect negative correlation, confirming Okun's Law. Output growth and unemployment changes move in opposite directions.
GDP_growth vs. SP500_return (0.50): A moderate positive correlation, showing that the real economy and financial markets tend to move together.
GDP_growth vs. CPI_inflation (0.64): A strong positive correlation, suggesting that periods of high growth are associated with rising inflation (a demand-pull relationship).

Unemp_change vs. SP500_return (-0.46): When unemployment rises, financial markets tend to fall, reflecting a weak economy.

Dynamic Propagation (VAR Model): The Figure: Impulse Response provides a more dynamic view of propagation. It models the effect of a shock from one variable onto another over time.

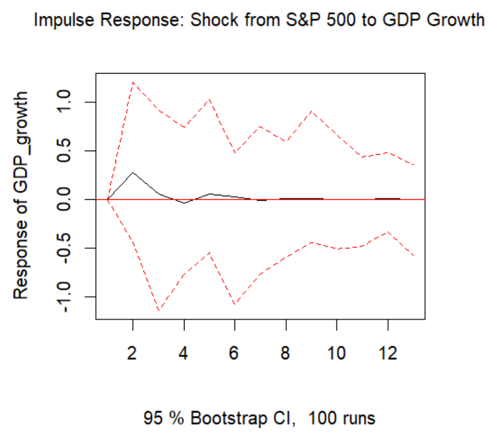

Figure 9: Impulse Response

Shock from SsP 500 to GDP Growth: The plot shows that a positive, one-standard-deviation shock to the SCP 500 (e.g., a sudden market rally) has a positive and statistically significant impact on Real GDP growth in the following quarter (quarter 2 on the x-axis).

Mechanism: This demonstrates a real amplification mechanism. A financial shock (either positive or negative) does not stay in the financial system. It propagates to the real economy, likely through wealth effects (consumers feel richer and spend more) and business confidence channels (firms see rising markets and increase investment). This mechanism was central to the GFC, where a *negative* financial shock from the housing market collapse propagated and amplified, crushing real GDP and employment.

2.5 Conclusion

The Global Financial Crisis and the COVID-19 recession were starkly different events. The GFC was an endogenous financial crisis that built for years, resulting in a severe, protracted recession with a slow, painful recovery. The COVID-19 recession was an exogenous, sudden-stop shock of unprecedented speed and depth. While its immediate impact was more severe, the recovery in output was (initially) equally rapid, creating a distinct "V-shape". The underlying economic mechanisms, such as Okun's Law and the propagation of financial shocks to the real economy, hold true in both, but the COVID-19 event pushed these relationships to an extreme never before seen.

3.Firm-Level Responses to the Global Financial Crisis

3.1 Objective and Data Framework

3.1.1 Understanding Compustat Data

To analyze how firms reacted to the Global Financial Crisis (GFC), this study uses the Compustat North America database, accessed via Wharton Research Data Services (WRDS). Compustat is the standard academic and financial industry source for firm-level data, providing comprehensive, standardized financial statement information for publicly traded companies in the United States and Canada. The database includes decades of quarterly and annual data from firms' income statements, balance sheets and cash flow statements, covering both active and inactive companies to prevent survivorship bias.

3.2 Key Variables and Relation to Business Cycles

To capture the multifaceted impact of the crisis, three key firm-level indicators were selected, as recommended by the project guidelines. These variables are constructed at a quarterly frequency to observe the crisis dynamics as they unfolded.

Revenue Growth (Demand-Side Impact): This indicator is calculated as the percentage change in a firm's quarterly sales compared to the same quarter in the previous year. Revenue is a primary measure of business activity and is strongly procyclical. It is expected to rise during economic expansions and fall sharply during recessions as consumer and business demand contracts.

Investment Spending (Capital Expenditure Cutbacks): Measured as quarterly capital expenditures as a percentage of a firm's total assets. Investment is one of the most volatile components of GDP and is also highly procyclical. Firms invest in new projects and equipment when they are confident about future demand. During a crisis, uncertainty and tight credit conditions are expected to cause significant capital expenditure cutbacks.

Leverage Ratios (Financial Stress): This is the ratio of a firm's total liabilities to its total assets, a key measure of financial stress. Its cyclicality is complex. While firms might take on more debt during booms, this ratio can spike countercyclically during a financial crisis. This is not necessarily because firms are borrowing more, but because the value of their assets (the denominator) collapses, mechanically increasing the ratio and pushing firms closer to default.

3.3 Data Panel and Analytical Approach

A firm-level panel dataset was constructed using the Compustat North America quarterly fundamentals and company files. The sample period covers 2005 through 2012, allowing for a clear analysis of firm performance "before," "during," and "after" the GFC. Based on NBER classifications and the provided figures, the GFC period is defined as 2007Q4 (the fourth quarter of 2007) through 2009Q2 (the second quarter of 2009).

To analyze aggregate trends, the cross-sectional median of each variable was calculated for each quarter. The median is used to provide a robust measure of the typical firm, as it is less sensitive to extreme outliers than the mean. For the sectoral analysis, firms were classified into "Manufacturing" (SIC codes 20-39) and "Retail" (SIC codes 52-59) and their average investment rates were compared. Finally, a regression analysis was performed to explain the observed patterns.

3.4 Analysis of Firm-Level Responses

The first set of figures plots the aggregate time series for the median firm, with the GFC period shaded in gray.

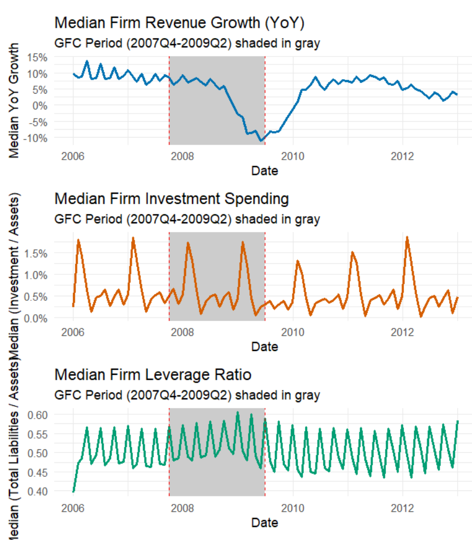

Figure 10: Median of a key firm-level variables (Revenue Growth, Investment Spending, Leverage Ratios)

The top panel shows that median year-over-year revenue growth was healthy and positive, fluctuating between 5% and 10% in the pre-crisis period. Coinciding with the start of the GFC in late 2007, revenue growth began a sharp and sustained decline, demonstrating the severe demand-side shock of the crisis. It plummeted to a trough of approximately -10% by early 2009. This signifies that the typical firm saw its sales fall 10% below the previous year's level. The recovery was V-shaped, with revenue growth quickly rebounding to positive territory by 2010 and stabilizing.

The middle panel, showing median investment, reveals strong seasonal patterns, with investment (as a percentage of assets) spiking at the end or beginning of each year. During the GFC, this seasonal pattern was visibly suppressed. The 2008 and 2009 peaks were lower than in 2006 or 2007 and the baseline level of investment fell. This provides clear evidence of widespread capital expenditure cutbacks as firms conserved cash and postponed projects in the face of extreme uncertainty.

The bottom panel illustrates the rise in financial stress. In the pre-crisis period, the median leverage ratio was stable at around 0.50 (or 50%). As the GFC unfolded, the median leverage ratio spiked, rising to nearly 0.60 (or 60%). This increase is a classic example of a crisis amplification mechanism. As the financial crisis tanked the value of firms' assets (such as real estate, securities and goodwill), the denominator of the leverage ratio fell, mechanically increasing the ratio even without new borrowing. This financial distress forced firms to cut investment and hiring, thus worsening the recession.

3.5 Sectoral Differences in Crisis Response

The second figure isolates investment spending to identify sectoral differences, comparing Manufacturing and Retail firms across the three periods. This analysis reveals significant heterogeneity.

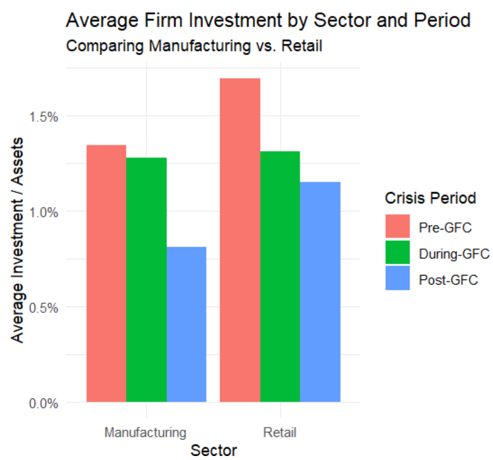

Figure 11: Average Firm Investment by Sector and Crisis Period

In the Pre-GFC period, Retail firms had a higher average investment rate (approx. 1.8% of assets) than Manufacturing firms (1.3%).

During the GFC, both sectors cut back, but the response differed. Retail firms, which are highly sensitive to immediate consumer demand, cut investment more sharply, with their average rate falling from 1.8% to 1.3%. Manufacturing investment, by contrast, remained relatively stable, dipping only slightly from 1.3% to 1.25%.

The most dramatic finding is in the Post-GFC period. Manufacturing investment, which was stable during the crisis, collapsed after the crisis, falling to 0.8%. This suggests that the impact on capital-intensive sectors was delayed but more persistent, likely due to long-term demand uncertainty and the difficulties of financing large projects in the post-crisis credit environment. Retail, in contrast, stabilized at a "new normal" of 1.15%, a lower rate than pre-crisis but still higher than manufacturing. This demonstrates that different sectors experienced the crisis and its aftermath on different timelines.

3.6 Discussing Explanations for Observed Patterns

The final figure attempts to explain why some firms cut investment more than others. It plots the coefficients from a regression where the dependent variable is the change in a firm's investment rate (comparing their average during the GFC to their average before the GFC). The plot shows the effect of two pre-crisis firm characteristics on this change.

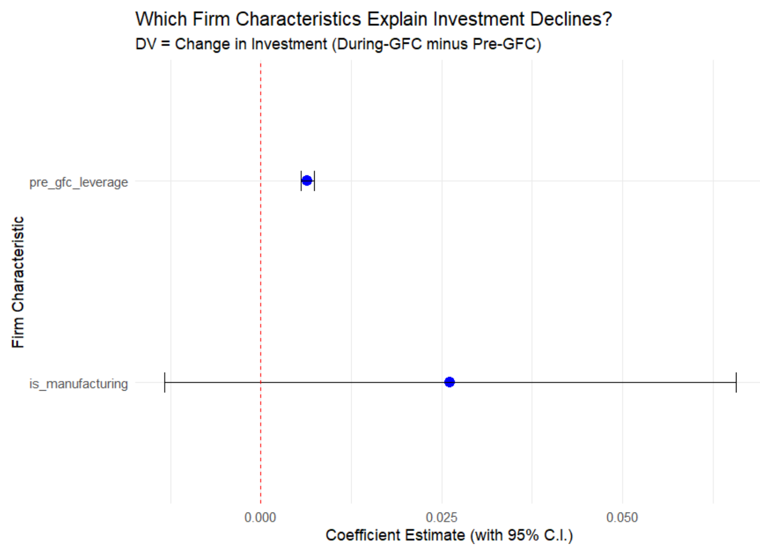

Figure 12: the change in a firm's investment rate (comparing their average during the GFC to their average before the GFC)

Manufacturing Sector: The coefficient for manufacturing is positive (approx. 0.025) and its 95% confidence interval does not cross zero, indicating a statistically significant result. This means that, all else equal, being in the manufacturing sector was associated with a smaller decline in investment during the GFC compared to the baseline of other sectors. This supports the finding from the sectoral plot, which showed Manufacturing investment was relatively stable during the crisis, with its main decline coming after.

Pre-GFC Leverage: The coefficient for pre-gfc_leverage is also positive (approx. 0.01) and statistically significant. This is a surprising and counter-intuitive finding. It suggests that firms with higher leverage before the crisis actually cut their investment less than low-leverage firms. This contradicts the common hypothesis that highly leveraged firms are riskier and face more credit constraints, which should force them to cut investment more. This puzzling result indicates that the drivers of investment cuts were more complex than pre-crisis leverage alone. It may be that the universal collapse in demand was such a dominant factor that it overshadowed typical financial constraints, or that the type of leverage (e.g., long-term bonds vs. short-term bank debt) mattered more than the simple ratio.